CODE SPORT

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In this month's column, we discuss some of the answers to last month's questions on machine learning.

Last month, we had discussed a bunch of interview questions on machine learning. While I had received answers to some of the questions from our readers, I also got feedback that the questions were complex and hence, it would be good to discuss the background of some of the topics related to these questions. This will enable readers to answer the more difficult questions. So, in this month's column, we will discuss some of the underlying concepts related to these questions without providing the canned solutions to them.

In last month's column, we had featured a couple of questions on auto-encoders. As you know, auto-encoders are a form of unsupervised learning technique. They are used to learn a representation of the input, which can then be fed to other tasks. This is typically known as pre-training, a topic that we will discuss later. Auto-encoders are typically used to compress a given input into a lower dimensional space, which can then be fed to a classification task. As in the case of PCA, auto-encoders are based on the premise that there is a certain correlation between some of the input dimensions, and so we can compress the input data into a fewer number of dimensions. However, if there is no correlation between the input dimensions, then compression into lower dimensions would not be possible and, hence, auto-encoders would not be able to learn a good compressed representation of the input.

A simple auto-encoder network is typically three-layered, with an input layer, a hidden layer whose dimensions are smaller than the input layer, and an output layer. Recall that the output from an auto-encoder is expected to be a faithful reproduction of the input. Here is a question to our readers: What is the dimensionality of the output layer? Well, as you may have guessed, it is the same as the input layer. Let us assume that we have an input image whose dimensions are 10x10 (100 pixels). Now, if we build an auto-encoder with an intermediate hidden layer of 50 units, the output of the hidden layer is a vector of dimension 50, which is a compressed version of the input. Basically, the auto-encoder understands the structure of the data (the correlated dimensions) and builds a representation of smaller dimensions.

On the other hand, let us assume that we have chosen a hidden layer of size 200. Would it still be possible to build a better representation of the input? The answer is: "Yes". If there is some interesting structural correlation in the data, by imposing certain constraints on the learnt representation, it is possible to get an improved representation, though it is of higher dimensions. For instance, one constraint that typically gets imposed is the sparsity constraint. We had discussed the idea behind sparse auto-encoders in one of our earlier columns. For more details, the reader can refer to the discussion in *http://ufldl.stanford.edu/tutorial/unsupervised/Autoencoders/*.

Last month's column had also raised the issue of whether to use Principal Component Analysis (PCA) for dimensionality reduction or use an auto-encoder for dimensionality reduction. The reader was asked to reason why one would use auto-encoder rather than PCA for dimensionality reduction. Let us assume that the non-linear activation function in the auto-encoder neural network was replaced by means of a linear function. Now consider the compressed representations created by PCA and auto-encoder, and whether they would be in a similar sub-space or not. Once you replace the linear activation function with a non-linear activation function in the auto-encoders, the input representations